	Scotch Student ID #			
Circle the relevant digits	0	0	0	0
	1	1	1	1
	2	2	2	2
	3	3	3	3
	4	4	4	4
	5	5	5	5
	6	6	6	6
	7	7	7	7
	8	8	8	8
	9	9	9	9

Teacher's Name

Scotch College

CHEMISTRY

SAC 2 – RATES, REDOX & EQUILIBRIUM

Date: Wednesday, May 25th

Reading Time	0 minutes
Writing Time	45 minutes

Task Sections	Marks	Your Marks
Total Marks	46	

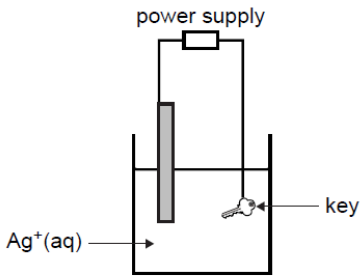
General Instructions
• Write your student number in the space provided above on this page. • Please complete all questions in the spaces provided.
Allowed Materials
• A VCAA Chemistry Data Book has been provided. • Students are permitted to bring into the SAC: pens, pencils, highlighters, erasers, sharpeners, rulers and one scientific calculator.
At the end of the task…
• Please ensure your student number (not name) has been filled in at the top of the page and that you have written your teacher's name.
Electronic Devices
Students are **not** allowed to have a mobile phone, smart watch and/or any other unauthorised electronic device in the SAC, unless it is TURNED OFF and is placed on the front teacher desk.

Q1. Which of the following pairs will react spontaneously?

I. $Sn(s) + Fe^{2+}(aq)$
II. $Cd(s) + Pb^{2+}(aq)$
III. $Fe(s) + Pb^{2+}(aq)$

A. I only
B. II only
C. III only
D. II and III only

Q2. A student decided to silver-plate a locker key using the apparatus shown.



In this cell, the key is the

A. anode and is connected to the positive terminal of the power supply.

B. anode and is connected to the negative terminal of the power supply.

C. cathode and is connected to the positive terminal of the power supply.

D. cathode and is connected to the negative terminal of the power supply.

Q3. Which statements are correct about the action of a catalyst in a chemical reaction?

I. It increases the energy of each collision.
II. It alters the pathway of the reaction to lower the Activation energy.
III. It remains unchanged at the end of the reaction.

A. I and II only

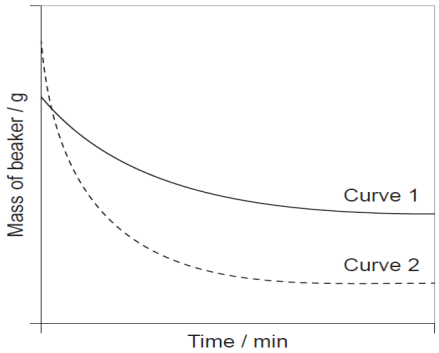
B. I and III only

C. II and III only

D. I, II and III

Q4. Curve 1 shows the mass change when marble chips are added to a ***large excess*** of hydrochloric acid in an open beaker.

$$CaCO_3\,(s) + 2HCl\,(aq) \rightarrow CaCl_2\,(aq) + H_2O(l) + CO_2\,(g)$$



Which changes would produce curve 2?

A. Powdering the same mass of marble chips and heating

B. Powdering the marble chips and doubling their mass

C. Doubling the volume of acid and heating

D. Doubling the acid concentration and powdering the same mass of marble chips

Q5. Hydrogen (H_2), a very useful fuel, can be produced from methane (CH_4) by reaction with steam:

$$CH_4\,(g) + H_2O\,(g) \rightleftharpoons CO\,(g) + 3H_2\,(g)$$

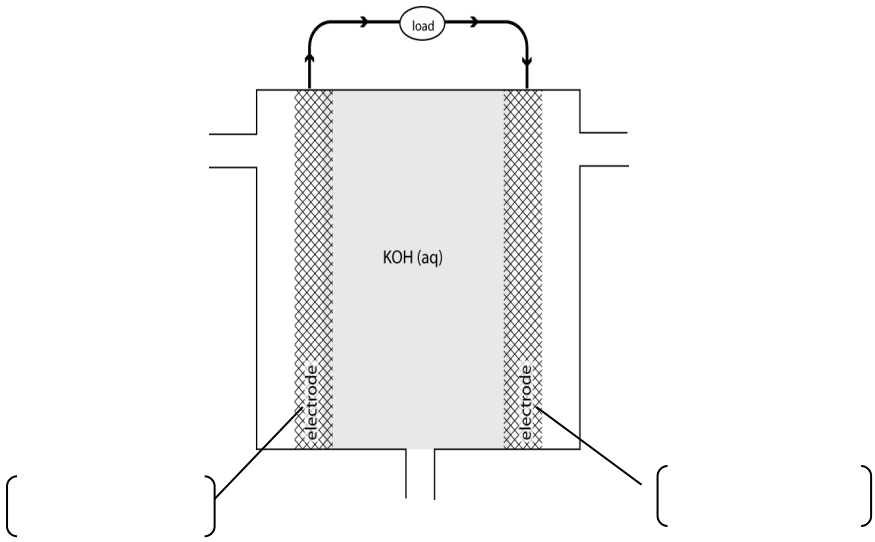
Conducting the reaction at atmospheric pressure, with a 50:50 mixture of CH_4 and H_2O, gave a disappointing equilibrium yield of hydrogen.

Which changes will produce the greatest increase in the percentage yield of methane into hydrogen at equilibrium?

	Pressure	**Proportion of H_2O (g)**
A.	Doubled	Halved
B.	Doubled	Doubled
C.	Halved	Doubled
D.	Halved	Halved

Q6. (3 marks)

The diagram below shows a version of an alkaline hydrogen-oxygen fuel cell. The arrows show the direction of electron flow in the external circuit.



(a) Add the labels ***anode*** and ***cathode*** in the appropriate spaces provided on the diagram above. 1 mark

(b) Add annotated labels to the diagram to show where the hydrogen gas and oxygen gas, respectively, are introduced. 1 mark

(c) How is this fuel cell different to other galvanic cells? 1 mark

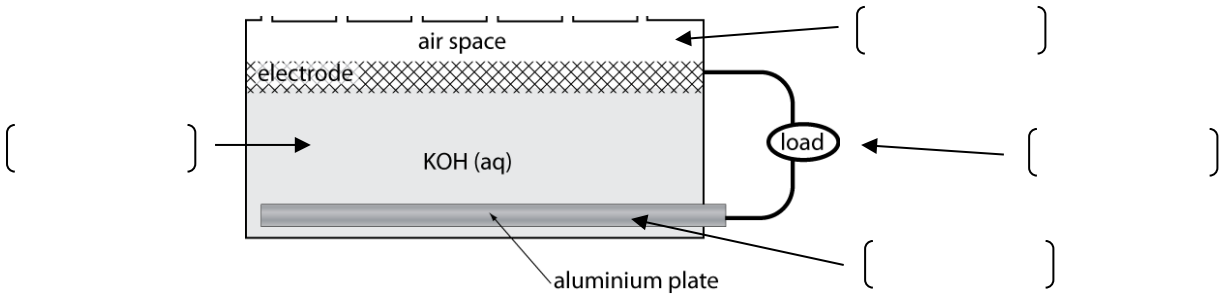
Q7. (6 marks)

The diagram below shows a version of an alkaline aluminium-air cell.

The half-reactions occurring are:

$$Al(s) \rightarrow Al^{3+}(aq) + 3e^-$$

$$O_2(g) + 2H_2O(l) + 4e^- \rightarrow 4OH^-(aq)$$



(a) Show the direction of electron flow through the external circuit on the diagram. 1 mark

(b) Label the **anode** in **correct** box above. 1 mark

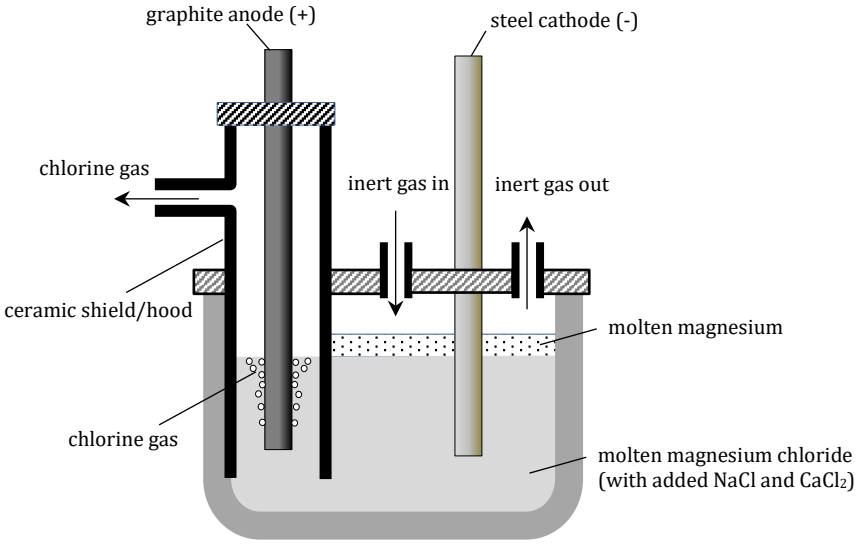
(c) Write an overall cell equation based on the two half-equations given, include states. 2 marks

(d) Using a nickel (Ni) plate in place of the aluminium plate gave a cell that still functioned. However, when a silver (Ag) plate was used instead the cell did not function.

Account for this difference by referring to the relative positions of the oxidants and reductants involved in the Electrochemical series. 2 marks

Q8. (7 marks)

Magnesium is one of the most abundant elements on Earth. It is used extensively in the production of magnesium-aluminium alloys It is produced by the electrolysis of molten magnesium chloride. A schematic diagram of the electrolytic cell is shown here.



The design of this cell takes into account the following properties of both magnesium metal and magnesium chloride:

• Molten magnesium reacts vigorously with oxygen.

• At the temperature of molten magnesium chloride, magnesium is a liquid.

• Molten magnesium has a lower density than molten magnesium chloride and forms a separate layer on the surface.

(a) Write a balanced half-equation for the reaction occurring at each of:

• the cathode 1 mark

• the anode. 1 mark

(b) Explain why an inert gas is constantly blown through the cathode compartment. 1 mark

(c) The melting point of a compound can often be lowered by the addition of small amounts of other compounds. In an industrial process, this will save energy. In this cell, NaCl and $CaCl_2$ are used to lower the melting point of $MgCl_2$. Why can NaCl and $CaCl_2$ be used to lower the melting point of $MgCl_2$ but $ZnCl_2$ cannot be used? 2 marks

(d) What difference would it make to the half-cell reactions if the graphite anode were replaced with an iron anode? Write the half-equation for any different half-cell reaction. Justify your answer. 2 marks

Q9. (13 marks)

To investigate the factors affecting the position of equilibrium established between Co^{2+}, Cl^- and the complex ion $CoCl_4^{2-}$ in aqueous, 5 mL of cobalt (II) solution and 4 mL of saturated sodium chloride solution were added together in a test tube. The equilibrium to be examined is as follows:

$$Co^{2+}(aq) + 4Cl^-(aq) \rightleftharpoons CoCl_4^{2-}(aq)$$

(pink) (colourless) (blue)

On reaching equilibrium the resulting mixture was dark pink. On heating the mixture became more blue in colour.

(a) State two characteristics of a reaction system that become true as it 'reaches equilibrium'? 2 marks

_____

_____

_____

(b) (i) Is the reaction Exothermic or Endothermic? _____ 1 mark

(ii) Referring to ***the results*** of the experiment explain your answer above. 1 mark

_____

_____

_____

(c) If $AgNO_3$ (aq) is added to the system, removing some of the chloride ions from the system, what effect will the subsequent shift have on the system's colour? 1 mark

Circle answer: MORE PINK STAY THE SAME MORE BLUE

Further experimentation of the original equilibrium added chloride ions to it, the solution becomes more blue in colour.

(d) Explain why this change occurred. 2 marks

_____

_____

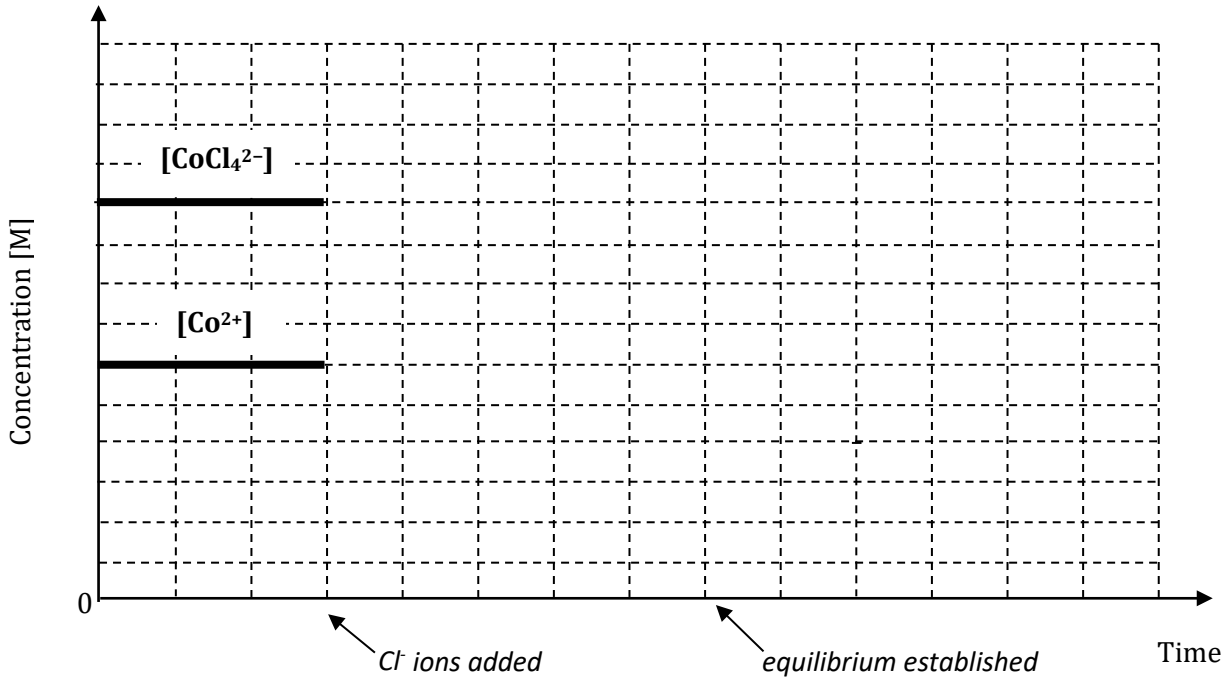
_____

(e) Explain the effect of the addition of chloride ions on the equilibrium constant, K. 1 mark

_____

(f) Show how the concentration of Co^{2+} and $CoCl_4^{2-}$ changed over time as Cl^- are added. 2 marks

Changes need not be drawn accurately to scale.



If water was added to dilute the original equilibrium mixture, with the temperature remaining constant:

$$Co^{2+}(aq) + 4Cl^-(aq) \rightleftharpoons CoCl_4^{2-}(aq)$$

(pink) (colourless) (blue)

(g) In which direction would equilibrium shift? _____ 1 mark

(h) Explain, using Le Chatelier's Principle, why the above change in the equilibrium system occurred. 2 marks

_____

_____

_____

_____

_____

Q10. (4 marks)

5.00 mol of N_2 and 6.00 mol of H_2 were mixed in an empty 4.00 L vessel at a specific temperature and reacted according to the equation:

$$N_2\,(g) + 3H_2\,(g) \rightleftharpoons 2NH_3\,(g)$$

When equilibrium was reached, it was found that 3.20 mol of NH_3 had been formed. Calculate the value of the equilibrium constant at this temperature. 4 marks

Q11. (8 marks)

Dimethyl ether, CH_3OCH_3, is used as an environmentally friendly propellant in spray cans. It can be synthesized from methanol according to the following equation.

$$2CH_3OH(g) \rightleftharpoons CH_3OCH_3(g) + H_2O(g) \qquad \Delta H = -24\ \text{kJ mol}^{-1}$$

The equilibrium constant, *K*, for this reaction at 350°C is 5.74.

(a) Write an expression for *K* for this reaction. 1 mark

(b) Calculate the value of *K* at 350°C for the following reaction. 1 mark

$$CH_3OCH_3(g) + H_2O(g) \rightleftharpoons 2CH_3OH(g)$$

(c) Methanol is pumped into an empty 20.0 L reactor vessel. At equilibrium the vessel contains 0.340 mol of methanol at 350°C. The concentration of the dimethyl ether and the water will the same as each other.

i. Calculate the concentration, in mol L^{-1}, of methanol at equilibrium. 1 mark

ii. Calculate the concentration, in mol L^{-1}, of dimethyl ether present at equilibrium**.** 2 marks

iii Calculate the amount, in mol, of dimethyl ether present at equilibrium. 1 mark

iii. Calculate the amount, in mol, of methanol initially pumped into the reaction vessel. 2 marks

END OF ASSESSMENT